



Modelling Financial Markets

Seminar 18: Systemic Risk and Networks — Hands-on



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Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BET-NG: Bucharest Exchange Trading – Energy and Related Utilities (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DAX: Deutscher Aktienindex (the German stock index (Frankfurt Stock Exchange))
- ▣ DBK: Deutsche Bank AG (Xetra ticker)
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ES: Expected Shortfall
- ▣ FEVD: Forecast Error Variance Decomposition
- ▣ FRM: Financial Risk Meter
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LRMES: Long-Run Marginal Expected Shortfall
- ▣ MES: Marginal Expected Shortfall
- ▣ OLS: Ordinary Least Squares
- ▣ QML: Quasi-Maximum Likelihood

Acronyms used in this material (2/2)

- ▣ **ROBOR: Romanian Interbank Offer Rate**
- ▣ **SG: Société Générale (the French banking group Société Générale)**
- ▣ **SRISK: Systemic RISK (capital shortfall measure)**
- ▣ **US: United States**
- ▣ **VAR: Vector AutoRegression**
- ▣ **VIX: CBOE Volatility IndeX**

Outline

Introduction

Part A: Pen and Paper

Part B: Data, Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive CoVaR, the Euler property of MES, the generalized FEVD and the reverse stress scenario
- ▣ Estimate LRMES and SRISK as published (GJR–GARCH, DCC, simulation)
- ▣ Test differences in ΔCoVaR on joint bootstrap draws, with multiplicity control
- ▣ Put intervals on a connectedness table and backtest a systemic measure

- ▣ Background: Lecture 18; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** ten computational exercises with inference (B1–B10)
- ▣ **Part C:** two open questions, seeds for the team project (C1, C3), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every exercise states the data and assumptions, the tasks (a), (b), . . . , what to report, and ends with an interpretation question
- Conventions: daily log returns in %; the level α is a tail probability; VaR, CoVaR and MES are positive losses
- The system of a bank is the equal-weighted portfolio of the **other** banks of its region

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (CoVaR under the Normal distribution), A3 (MES as an Euler contribution), A5 (the generalized FEVD), A7 (reverse stress test by a Lagrangian)
- Part B: B1 (tests for ΔCoVaR of US banks), B3 (LRMES by GJR–GARCH/DCC simulation), B5 (sensitivity of the connectedness index), B7 (the FRM penalty in one window), B9 (bootstrap intervals for connectedness)

Proposed: you solve them

- Part A: A2 (Girardi–Ergün CoVaR; model: A1), A4 (the 18 in LRMES; model: A3), A6 (bivariate connectedness; model: A5), A8 (Student- t reverse stress; model: A7)
- Part B: B2 (European banks, VaR ranking; model: B1), B4 (SRISK of DBK and TLV; model: B3), B6 (connectedness after 2020; model: B5, B9), B8 (Romanian stress test; model: A7), B10 (backtest of CoVaR; model: B1, A2)
- Part C: C1, open question and project seed; C2, critique an AI answer on CoVaR, MES and SRISK (model: A1, A3, Lecture 18); C3, the Romanian Financial Risk Meter out of sample (model: B7, Lecture 18 case study)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter18_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `scikit-learn`, `matplotlib`
- Series: six US banks, five European banks, Banca Transilvania and BRD; S&P 500, Euro Stoxx 50, BET, VIX; daily prices January 2010 – 18 September 2026
- Prices joined on common trading days first, then log returns; Romanian banks without the days with no trades

Formulas Used Today

- ▣ Quantile regression at level α : $q_\alpha(X^{\text{sys}} | X_i) = a_\alpha + b_\alpha X_i$; $\text{CoVaR}_\alpha = -(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i))$,
 $\Delta \text{CoVaR}_\alpha = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$ [Adrian & Brunnermeier \(2016\)](#)
- ▣ $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i | X_m \leq q_\alpha(X_m)]$ [Acharya et al. \(2017\)](#); approximation $\text{LRMES} \approx 1 - \exp(-18 \text{MES}^2\%)$ [Acharya, Engle & Richardson \(2012\)](#)
- ▣ $\text{SRISK} = kD - (1 - k)(1 - \text{LRMES})W$; break-even leverage $L^* = 1 + (1 - k)(1 - \text{LRMES})/k$ [Brownlees & Engle \(2017\)](#)
- ▣ Connectedness [Diebold & Yilmaz \(2014\)](#): $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$, $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$, net $T_i - F_i$, total $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$
- ▣ Reverse stress test: $f^* = -\ell^* \Sigma b / (b^\top \Sigma b)$, $d^* = \ell^* / \sqrt{b^\top \Sigma b}$

A1: Deriving CoVaR under the Normal Distribution [Solved]

Task

□ Data and assumptions

- ▶ (X_i, X^{sys}) : daily returns (%) of bank i and of the system, bivariate Normal with zero means, standard deviations σ_i , σ_{sys} and correlation $\rho > 0$
- ▶ Distress of the bank: $X_i = q_\alpha(X_i) = \sigma_i z_\alpha$, with $z_\alpha = \Phi^{-1}(\alpha) < 0$; median state: $X_i = 0$; CoVaR is minus the α -quantile of X^{sys} , a positive loss [Adrian & Brunnermeier \(2016\)](#)
- ▶ Numbers for (c): $\sigma_i = 2\%$, $\sigma_{sys} = 1.5\%$, $\rho = 0.6$, $\alpha = 1\%$

□ Tasks

- ▶ (a) Project X^{sys} on X_i and derive the conditional distribution of X^{sys} given $X_i = x$ (mean and variance).
- ▶ (b) Derive CoVaR_α (given $X_i = \sigma_i z_\alpha$), the median-state CoVaR (given $X_i = 0$) and $\Delta\text{CoVaR}_\alpha$ as their difference; show that $\Delta\text{CoVaR}_\alpha$ does not depend on σ_i .
- ▶ (c) Compute VaR 1% of the bank, CoVaR 1%, the median-state CoVaR and ΔCoVaR 1% for the numbers above.

□ Report

- ▶ The conditional mean and standard deviation, the three closed forms of (b) and the four numbers of (c), with every intermediate product

□ Interpretation question

- ▶ A bank twice as volatile, with the same ρ , has the same ΔCoVaR : what does this say about ranking banks by their own VaR?

A1: Deriving CoVaR under the Normal Distribution (Solution 1/2) [Solved]

- (a) Projection: $X^{sys} = \beta X_i + U$ with $\beta = \text{Cov}(X^{sys}, X_i) / \sigma_i^2 = \rho \sigma_{sys} / \sigma_i$ and U Normal, independent of X_i .
- Residual variance: $\text{Var}(U) = \sigma_{sys}^2 - \beta^2 \sigma_i^2 = \sigma_{sys}^2 (1 - \rho^2)$.
- Result: $X^{sys} \mid X_i = x \sim N\left(\rho \frac{\sigma_{sys}}{\sigma_i} x, \sigma_{sys}^2 (1 - \rho^2)\right)$.
- (b) The α -quantile given x is $\rho \frac{\sigma_{sys}}{\sigma_i} x + \sigma_{sys} \sqrt{1 - \rho^2} z_\alpha$.
- At $x = \sigma_i z_\alpha$: $\text{CoVaR}_\alpha = -\sigma_{sys} z_\alpha (\rho + \sqrt{1 - \rho^2})$; at $x = 0$: $\text{CoVaR}_\alpha^{50\%} = -\sigma_{sys} \sqrt{1 - \rho^2} z_\alpha$.
- Difference: $\Delta \text{CoVaR}_\alpha = -\rho \sigma_{sys} z_\alpha$; σ_i cancels because the distress state x grows with σ_i while the slope falls with $1/\sigma_i$.

A1: Deriving CoVaR under the Normal Distribution (Solution 2/2) [Solved]

- (c) $z_{1\%} = -2.326$; VaR 1% of the bank: $-\sigma_i z_{1\%} = 2 \cdot 2.326 = 4.65\%$.
- Slope $\rho\sigma_{sys}/\sigma_i = 0.6 \cdot 1.5/2 = 0.45$; conditional standard deviation $1.5\sqrt{1 - 0.36} = 1.20\%$.
- CoVaR 1%: $0.45 \cdot 4.65 + 1.20 \cdot 2.326 = 2.094 + 2.792 = 4.885\%$.
- Median state: $1.20 \cdot 2.326 = 2.792\%$; so $\Delta\text{CoVaR}_{1\%} = 0.6 \cdot 1.5 \cdot 2.326 = 2.094\%$, the first term above.
- Interpretation: when the bank moves from its median to its VaR, the system loses about 2.09 extra points; doubling σ_i doubles the VaR but leaves ΔCoVaR unchanged.
- Own risk and contribution to systemic risk are different quantities: a ranking by VaR is not a systemic ranking.

A2: The Girardi–Ergün CoVaR under the Normal Distribution (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Same bivariate Normal model and numbers as A1(c): $\sigma_{sys} = 1.5\%$, $\rho = 0.6$, $\alpha = 1\%$
- ▶ Definition of [Girardi & Ergün \(2013\)](#): distress is $X_i \leq q_\alpha(X_i)$ (at or beyond the VaR), and $\text{CoVaR}_\alpha^\leq = c$ solves $P(X^{sys} \leq -c \mid X_i \leq q_\alpha(X_i)) = \alpha$
- ▶ $\Phi_2(u, v; \rho)$: the standard bivariate Normal cdf with correlation ρ

□ Tasks

- ▶ (a) Write the conditional probability as a joint probability divided by α , standardise both coordinates and show that c solves $\Phi_2(z_\alpha, -c/\sigma_{sys}; \rho) = \alpha^2$.
- ▶ (b) Prove that $\text{CoVaR}_\alpha^\leq \geq \text{CoVaR}_\alpha$ for $\rho > 0$, using that the law of X^{sys} given $X_i \leq q_\alpha$ is a mixture, over $x \leq q_\alpha$, of the laws of A1(a).
- ▶ (c) Solve the equation of (a) numerically by bisection on c and compare the result with $\text{CoVaR}_{1\%}$ of A1.

A2: The Girardi–Ergün CoVaR under the Normal Distribution (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ c , the standardised threshold $-c/\sigma_{sys}$ and the gap to A1 in points and in percent
- ▶ Model: A1

▣ Interpretation question

- ▶ The gap is small under the Normal distribution: why would it be much larger in bank data?

A3: MES as an Euler Contribution [Solved]

Task

□ Data and assumptions

- ▶ Portfolio $X_p = \sum_i w_i X_i$ of daily returns with a continuous distribution; $q_\alpha(w)$ is its α -quantile
- ▶ $ES_\alpha(X_p) = -\frac{1}{\alpha} \mathbb{E}[X_p \mathbf{1}\{X_p \leq q_\alpha\}]$ and $MES_{i,\alpha} = -\mathbb{E}[X_i \mid X_p \leq q_\alpha]$ [Acharya et al. \(2017\)](#)
- ▶ Numbers for (c): jointly Normal, zero means, $w = (0.5, 0.5)$, $\sigma = (2\%, 3\%)$, $\rho = 0.5$, $\alpha = 5\%$

□ Tasks

- ▶ (a) Differentiate ES_α with respect to w_i , including the effect of w_i on the quantile, to show $\partial ES_\alpha / \partial w_i = MES_{i,\alpha}$; then use homogeneity to show $ES_\alpha = \sum_i w_i MES_{i,\alpha}$.
- ▶ (b) For jointly Normal returns with zero means, show that $MES_{i,\alpha} = \beta_{i,p} ES_\alpha(X_p)$ with $\beta_{i,p} = \text{Cov}(X_i, X_p) / \sigma_p^2$ and $ES_\alpha(X_p) = \sigma_p \varphi(z_\alpha) / \alpha$.
- ▶ (c) Compute σ_p , $\varphi(z_{5\%})$, both betas and both MES for the numbers above, and check the Euler sum.

□ Report

- ▶ σ_p , $\varphi(z_{5\%})/0.05$, $ES_{5\%}$, β , MES and the check $\sum_i w_i MES_i = ES_{5\%}$

□ Interpretation question

- ▶ Asset 2 has half the total portfolio weight: what share of the ES does it carry, and why is MES, not volatility, the right key for allocating tail risk?

A3: MES as an Euler Contribution (Solution 1/2) [Solved]

- (a) Write $\alpha \text{ES}_\alpha = -\mathbb{E}[X_p \mathbf{1}\{X_p \leq q(w)\}]$; differentiating under the expectation gives the direct term $-\mathbb{E}[X_i \mathbf{1}\{X_p \leq q\}]$.
- The boundary term is $-q \frac{\partial}{\partial w_i} P(X_p \leq q(w))$: the mass that crosses the boundary, where $X_p = q$.
- $P(X_p \leq q(w)) = \alpha$ for every w , so this derivative is zero and $\partial \text{ES}_\alpha / \partial w_i = -\frac{1}{\alpha} \mathbb{E}[X_i \mathbf{1}\{X_p \leq q\}] = \text{MES}_{i,\alpha}$.
- Homogeneity: $\text{ES}_\alpha(tX_p) = t \text{ES}_\alpha(X_p)$, so Euler's theorem gives $\text{ES}_\alpha = \sum_i w_i \partial \text{ES}_\alpha / \partial w_i = \sum_i w_i \text{MES}_{i,\alpha}$ [Gouriéroux, Laurent & Scaillet \(2000\)](#).
- (b) Normal regression: $\mathbb{E}[X_i | X_p] = \beta_{i,p} X_p$, so $\text{MES}_{i,\alpha} = -\beta_{i,p} \mathbb{E}[X_p | X_p \leq q] = \beta_{i,p} \text{ES}_\alpha(X_p)$.
- For $X_p \sim N(0, \sigma_p^2)$, $\text{ES}_\alpha = \sigma_p \varphi(z_\alpha) / \alpha$; and $\sum_i w_i \beta_{i,p} = w^\top \Sigma w / \sigma_p^2 = 1$ confirms the Euler sum.

A3: MES as an Euler Contribution (Solution 2/2) [Solved]

- (c) Covariance $\rho\sigma_1\sigma_2 = 0.5 \cdot 2 \cdot 3 = 3$; $\sigma_p^2 = 0.25 \cdot 4 + 2 \cdot 0.25 \cdot 3 + 0.25 \cdot 9 = 4.75$, so $\sigma_p = 2.179\%$.
- $\varphi(z_{5\%})/0.05 = 2.063$, so $ES_{5\%} = 2.179 \cdot 2.063 = 4.50\%$.
- $\text{Cov}(X_i, X_p) = (\Sigma w)_i = (3.5, 6.0)$; $\beta = (3.5, 6.0)/4.75 = (0.737, 1.263)$.
- $MES = \beta \cdot 4.50 = (3.31, 5.68)\%$; check: $0.5(3.31 + 5.68) = 4.50\% = ES_{5\%}$.
- Interpretation: with half the weight, asset 2 carries 63% of the ES, because it moves more with the portfolio in its tail.
- MES splits the tail loss additively and uses the co-movement with the portfolio, which volatility alone ignores.

A4: Where the 18 in the LRMES Shortcut Comes From (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Crisis scenario [Acharya, Engle & Richardson \(2012\)](#): the market falls by exactly $d = 40\%$ over six months, a deterministic boundary scenario ($R_m = -d$) that motivates the shortcut; the simulated LRMES of B3 instead averages over all paths with $R_m < C$
- ▶ The bank's six-month log return is β times the market's, with constant β ; $\text{MES}^{2\%} = \beta \text{ES}_m^{2\%}$, where $\text{ES}_m^{2\%}$ is the average market loss on days with a return below -2%
- ▶ Data for (c): S&P 500 daily log returns (GSPC.INDX, close), 5 January 2010 – 18 September 2026; and a Normal market with $\sigma_m = 1\%$

□ Tasks

- ▶ (a) Show that in this scenario the bank loses $\text{LRMES} = 1 - (1 - d)^\beta$.
- ▶ (b) Substitute $\beta = \text{MES}^{2\%} / \text{ES}_m^{2\%}$ to show $\text{LRMES} = 1 - \exp(-\kappa \text{MES}^{2\%})$ with $\kappa = -\ln(1 - d) / \text{ES}_m^{2\%}$.
- ▶ (c) Compute $\text{ES}_m^{2\%}$ and κ for the S&P 500 and for the Normal market ($\text{ES}_m^{2\%} = \sigma_m \varphi(2) / \Phi(-2)$), and the $\text{ES}_m^{2\%}$ that gives $\kappa = 18$.

A4: Where the 18 in the LRMES Shortcut Comes From (2/2) [Proposed]

Solution discussed in class

Task

□ Report

- ▶ The number of S&P 500 days below -2% , both values of $ES_m^{2\%}$ and of κ , and the $ES_m^{2\%}$ implied by 18
- ▶ Model: A3

□ Interpretation question

- ▶ What goes wrong in the shortcut if β rises in crises?

A5: Deriving the Generalized FEVD (1/2) [Solved]

Task

□ Data and assumptions

- ▶ Stable VAR(1) $y_t = Ay_{t-1} + \varepsilon_t$ (eigenvalues of A inside the unit circle), ε_t serially uncorrelated, zero-mean and jointly Normal with positive definite $\text{Cov}(\varepsilon_t) = \Sigma = (\sigma_{ij})$; moving-average matrices $\Phi_h = A^h$; e_j is the j -th unit vector
- ▶ Generalized impulse [Pesaran & Shin \(1998\)](#): a one-standard-deviation shock $\delta_j = \sqrt{\sigma_{jj}}$ to variable j , the other shocks at their conditional expectation
- ▶ Numbers for (b): $A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

□ Tasks

- ▶ (a) Show that $\mathbb{E}[\varepsilon_t \mid \varepsilon_{jt} = \delta_j] = \Sigma e_j / \sqrt{\sigma_{jj}}$ and derive the share $\theta_{ij}(H)$ of the H -step forecast error variance of i due to shocks in j .
- ▶ (b) Compute θ , its row sums, the row-normalised table $\tilde{\theta}$ and total connectedness C at $H = 1$ and $H = 2$.
- ▶ (c) Compute the Cholesky decomposition at $H = 1$ with variable 1 first, and show that with Σ diagonal the generalized and Cholesky decompositions coincide.

A5: Deriving the Generalized FEVD (2/2) [Solved]

Task

▣ Report

► θ and $\tilde{\theta}$ with row sums and C at $H = 1, 2$, with $A\Sigma$ and $\text{diag}(A\Sigma A^\top)$; the Cholesky table

▣ Interpretation question

► Why do the rows of the generalized table not sum to one, and what does the Cholesky order decide?



A5: Deriving the Generalized FEVD (Solution 1/3) [Solved]

- (a) For Normal shocks $\mathbb{E}[\varepsilon_t | \varepsilon_{jt}] = \Sigma e_j \varepsilon_{jt} / \sigma_{jj}$; at $\varepsilon_{jt} = \sqrt{\sigma_{jj}}$ this is $\Sigma e_j / \sqrt{\sigma_{jj}}$.
- Response of variable i after h steps: $e_i^\top \Phi_h \Sigma e_j / \sqrt{\sigma_{jj}}$; the H -step forecast error variance of i is $\sum_{h < H} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i$.
- Share: $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h < H} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h < H} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$.
- Normalisation $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$ and total connectedness $C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$ Diebold & Yilmaz (2014).

A5: Deriving the Generalized FEVD (Solution 2/3) [Solved]

- (b) $H = 1$ ($\Phi_0 = I$): $\theta_{ij} = \sigma_{ij}^2 / (\sigma_{ii}\sigma_{jj})$, so $\theta = \begin{pmatrix} 1 & 0.25 \\ 0.25 & 1 \end{pmatrix}$, row sums 1.25.
- Normalised: $\tilde{\theta} = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$, so $C = 100(0.2 + 0.2)/2 = 20.0\%$.
- $H = 2$ adds $\Phi_1 = A$: $A\Sigma = \begin{pmatrix} 0.60 & 0.45 \\ 0.30 & 0.45 \end{pmatrix}$, $\text{diag}(A\Sigma A^\top) = (0.39, 0.21)$; denominators 1.39 and 1.21.
- Numerators $\sigma_{ij}^2 + (A\Sigma)_{ij}^2$: row 1 (1.3600, 0.4525), row 2 (0.3400, 1.2025); so $\theta = \begin{pmatrix} 0.978 & 0.326 \\ 0.281 & 0.994 \end{pmatrix}$, row sums 1.304 and 1.275.
- Normalised off-diagonal shares 0.250 and 0.220, so $C = 100(0.250 + 0.220)/2 = 23.5\%$.

A5: Deriving the Generalized FEVD (Solution 3/3) [Solved]

- (c) Cholesky with variable 1 first: $P = \begin{pmatrix} 1 & 0 \\ 0.5 & \sqrt{0.75} \end{pmatrix}$, $PP^\top = \Sigma$; at $H = 1$ the shares are the squared entries of the rows of P : (1, 0) and (0.25, 0.75).
- The whole common shock is given to variable 1, the first in the order.
- Diagonal Σ : $\Sigma e_j / \sqrt{\sigma_{jj}} = \sqrt{\sigma_{jj}} e_j = P e_j$ with $P = \Sigma^{1/2}$ the diagonal Cholesky factor: the impulses coincide and the rows already sum to one.
- Interpretation: rows of θ exceed one because a correlated shock is counted for both variables; normalisation fixes the sum without imposing an order.
- With Cholesky the order, not the data, decides who receives the common shock; the lagged links of A raise C from 20.0% to 23.5% between $H = 1$ and $H = 2$.

A6: Connectedness of a Bivariate VAR [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ VAR(1) for two banks with $A = \begin{pmatrix} 0.5 & 0 \\ 0.4 & 0.5 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}$
- ▶ $A_{21} = 0.4$, $A_{12} = 0$: bank 1 drives bank 2 with a one-day lag, not the reverse
- ▶ Generalized decomposition of A5, rows normalised to one; $C = \frac{100}{2}(\tilde{\theta}_{12} + \tilde{\theta}_{21})$; net position $T_i - F_i$ (to others minus from others)

□ Tasks

- ▶ (a) Compute the generalized table (raw and normalised) and C at $H = 1$.
- ▶ (b) Compute $A\Sigma$, $\text{diag}(A\Sigma A^\top)$, the raw and normalised tables and C at $H = 2$.
- ▶ (c) Compute the net position of each bank at $H = 2$ and identify the net transmitter.

□ Report

- ▶ Both tables, C at both horizons and the two net positions
- ▶ Model: A5

□ Interpretation question

- ▶ Why is the table symmetric at $H = 1$ although bank 1 drives bank 2, and why does C rise with H ?

A7: Reverse Stress Test: the Lagrangian [Solved]

Task

□ Data and assumptions

- ▶ Two risk factors $f = (f_1, f_2)^\top \sim N(0, \Sigma)$ (four-week returns, %); portfolio loss $L = -b^\top f$ with exposures b
- ▶ Reverse stress test: the most plausible scenario (highest density) that produces a given loss ℓ^*
- ▶ Numbers for (c): $b = (1.2, 0.8)$, $\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ ($\%^2$), $\ell^* = 20\%$

□ Tasks

- ▶ (a) Minimise $\frac{1}{2}f^\top \Sigma^{-1}f$ subject to $-b^\top f = \ell^*$ with a Lagrangian and derive f^* .
- ▶ (b) Show that the Mahalanobis distance of f^* is $d^* = \ell^* / \sqrt{b^\top \Sigma b}$ and that $P(L \geq \ell^*) = \Phi(-d^*)$.
- ▶ (c) Compute Σb , $b^\top \Sigma b$, f^* , d^* and $P(L \geq \ell^*)$, and check that f^* produces the target loss.

□ Report

- ▶ Every intermediate vector and scalar, f^* , d^* and the probability

□ Interpretation question

- ▶ Which factor carries the larger fall in the scenario, and how should a risk committee read a probability of this size?

A7: Reverse Stress Test: the Lagrangian (Solution 1/2) [Solved]

- (a) Maximum density is minimum $\frac{1}{2}f^\top \Sigma^{-1}f$; Lagrangian $\mathcal{L} = \frac{1}{2}f^\top \Sigma^{-1}f + \lambda(b^\top f + \ell^*)$.
- First-order condition: $\Sigma^{-1}f + \lambda b = 0$, so $f = -\lambda \Sigma b$.
- Constraint: $-b^\top f = \lambda b^\top \Sigma b = \ell^*$, so $\lambda = \ell^*/(b^\top \Sigma b)$ and $f^* = -\ell^* \Sigma b/(b^\top \Sigma b)$.
- (b) $f^{*\top} \Sigma^{-1} f^* = \lambda^2 b^\top \Sigma b = \ell^{*2}/(b^\top \Sigma b)$, so $d^* = \ell^*/\sqrt{b^\top \Sigma b}$.
- $L = -b^\top f \sim N(0, b^\top \Sigma b)$, so $P(L \geq \ell^*) = 1 - \Phi(\ell^*/\sqrt{b^\top \Sigma b}) = \Phi(-d^*)$.

A7: Reverse Stress Test: the Lagrangian (Solution 2/2) [Solved]

- (c) $\Sigma b = (16 \cdot 1.2 + 6 \cdot 0.8, 6 \cdot 1.2 + 9 \cdot 0.8) = (24, 14.4)$.
- $b^\top \Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, $\sqrt{40.32} = 6.35$; $\lambda = 20/40.32 = 0.496$.
- $f^* = -0.496 \cdot (24, 14.4) = (-11.9\%, -7.1\%)$; check: $1.2 \cdot 11.90 + 0.8 \cdot 7.14 = 20\%$.
- $d^* = 20/6.35 = 3.15$ and $P(L \geq 20\%) = \Phi(-3.15) = 0.082\%$.
- Interpretation: factor 1 falls by 11.90% (2.98 of its standard deviations), factor 2 by 7.14% (2.38): the scenario leans on the factor with the larger exposure and variance.
- Under the Normal model such a four-week loss happens about once in 94 years; A8 shows how fragile this number is.

A8: Reverse Stress Test under a Student- t Distribution [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Same b and Σ as A7; targets $\ell^* = 20\%$ and $\ell^* = 30\%$
- ▶ f is multivariate Student- t with $\nu = 4$ degrees of freedom and covariance Σ , so its scale matrix is $\Sigma(\nu - 2)/\nu$; T_ν is the Student- t cdf

□ Tasks

- ▶ (a) Show that the most plausible scenario f^* is the same as in A7, using that the density depends on f only through $f^\top \Sigma^{-1} f$ and decreases in it.
- ▶ (b) Show that $L = -b^\top f$ is a univariate t_ν with scale $\sqrt{b^\top \Sigma b (\nu - 2)/\nu}$, so that $P(L \geq \ell^*) = 1 - T_\nu(d^* \sqrt{\nu/(\nu - 2)})$.
- ▶ (c) Compute the probability for both targets under the Normal and the t_4 models, and the ratio of the two.

□ Report

- ▶ d^* , the t argument, the two probabilities and their ratio for each target
- ▶ Model: A7

□ Interpretation question

- ▶ What is robust in a reverse stress test, and what is not?

B1: ΔCoVaR 5% of US Banks, with Tests (1/2) [Solved]

Task

□ Data and assumptions

- ▶ Adjusted closes of six US banks in data/market: JPM.US, BAC.US, C.US, GS.US, MS.US, WFC.US
- ▶ Sample: 5 January 2010 – 18 September 2026, 3 960 trading days common to all 13 banks and the four indices (prices joined first, then daily log returns in %)
- ▶ System of bank i : the equal-weighted portfolio of the other five banks [Adrian & Brunnermeier \(2016\)](#); level $\alpha = 5\%$, about 198 tail days

□ Tasks

- ▶ (a) For each bank, estimate the quantile regression $q_{5\%}(X_t^{\text{sys}} \mid X_{i,t}) = a_{5\%} + b_{5\%}X_{i,t}$ and compute VaR 5%, CoVaR 5% and $\Delta\text{CoVaR}_{5\%} = \hat{b}_{5\%}(\hat{q}_{50\%} - \hat{q}_{5\%})$.
- ▶ (b) Build 95% percentile intervals for ΔCoVaR with a moving-block bootstrap (blocks of 20 days) [Künsch \(1989\)](#) and with an i.i.d. bootstrap, 999 resamples each.
- ▶ (c) Test $H_0 : b_{5\%} = b_{50\%}$ for each bank with $z = (\hat{b}_{5\%} - \hat{b}_{50\%})/\text{se}$ (block-bootstrap se) at the 5% level, with and without the Holm correction [Holm \(1979\)](#).
- ▶ (d) Test equal ΔCoVaR for the 15 pairs on joint draws (the same resampled days for all banks) with Holm, and for all six banks jointly with a Wald test (χ^2_5).

B1: ΔCoVaR 5% of US Banks, with Tests (2/2) [Solved]

Task

□ Report

- ▶ A table with $\hat{b}_{5\%}$, ΔCoVaR , both intervals, $\hat{b}_{5\%} - \hat{b}_{50\%}$ and its p -value; a chart of the two intervals per bank
- ▶ The number of rejections in (c) and (d), raw and after Holm; the Wald statistic, its p -value and the decision

□ Interpretation questions

- ▶ Why is the block interval wider than the i.i.d. one?
- ▶ Can overlapping marginal intervals hide a significant difference?
- ▶ Why is bank i left out of its own system?

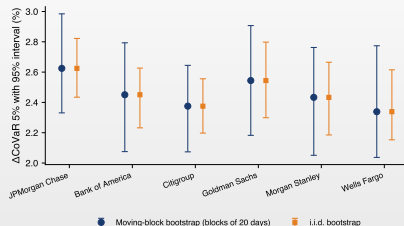
B1: ΔCoVaR 5% of US Banks, with Tests (Solution 1/3) [Solved]

- Step 1: for each bank the system return is $X_t^{\text{sys}} = \frac{1}{5} \sum_{j \neq i} X_{j,t}$; estimate the quantile regressions at 5% and at 50%.
- Step 2: plug in the empirical quantiles: $\text{CoVaR}_{5\%} = -(\hat{\alpha} + \hat{b}\hat{q}_{5\%})$ and $\Delta\text{CoVaR}_{5\%} = \hat{b}(\hat{q}_{50\%} - \hat{q}_{5\%})$.
- Example, JPMorgan: $\text{VaR } 5\% = 2.66\%$, $\hat{q}_{50\%} - \hat{q}_{5\%} = 2.74$, $\hat{b}_{5\%} = 0.96$, so $\Delta\text{CoVaR}_{5\%} = 0.96 \cdot 2.74 = 2.62\%$; $\text{CoVaR } 5\% = 3.81\%$.
- Step 3: draw 999 samples of 20-day blocks (block bootstrap) and 999 samples of single days (i.i.d.), re-estimate on each and take the 2.5% and 97.5% percentiles.
- Step 4: in each block draw compute all six $\hat{b}_{5\%} - \hat{b}_{50\%}$ and all six ΔCoVaR on the same days; the standard deviations of the draws are the standard errors, their covariance feeds the Wald test.

B1: ΔCoVaR 5% of US Banks, with Tests (Solution 2/3) [Solved]

	$\hat{b}_{5\%}$	ΔCoVaR	block 95% interval	i.i.d. 95% interval	$\hat{b}_{5\%} - \hat{b}_{50\%}$	p
JPMorgan	0.96	2.62	[2.33, 2.98]	[2.43, 2.82]	-0.002	0.878
Bank of America	0.76	2.45	[2.08, 2.79]	[2.23, 2.63]	-0.017	0.425
Citigroup	0.72	2.38	[2.07, 2.64]	[2.20, 2.56]	-0.040	0.019
Goldman Sachs	0.89	2.54	[2.18, 2.91]	[2.30, 2.80]	0.044	0.048
Morgan Stanley	0.72	2.43	[2.05, 2.76]	[2.19, 2.66]	-0.009	0.746
Wells Fargo	0.81	2.34	[2.04, 2.77]	[2.15, 2.61]	0.032	0.221

- (a) All six contributions lie between 2.34% and 2.62% of extra system loss; slopes lie between 0.72 and 0.96.
- (b) Block intervals are wider (JPMorgan: width 0.65 against 0.39 points): tail days cluster in crises, and the i.i.d. bootstrap breaks the clusters apart.

B1: ΔCoVaR 5% of US Banks, with Tests (Solution 3/3) [Solved]

- (c) Citigroup: $z = -0.040/0.017 = -2.34$, $p = 0.019$, rejected at 5%; JPMorgan: $z = -0.15$, $p = 0.878$, not rejected; overall, the null is rejected for 2 of 6 banks, and for 0 after Holm.
- (d) 2 of 15 pairs differ at 5%, 0 after Holm; Wald $\chi^2_5 = 14.2$, $p = 0.014 < 0.05$: we reject that the six contributions are all equal.
- Interpretation: the joint test sees differences that overlapping marginal intervals hide, because it uses the positive correlation of the estimates across banks.
- Bank i is left out of its system: otherwise X_i would appear on both sides of the regression and add a mechanical self-contribution to $\hat{b}$ (with weight w and slope b on the rest, the slope becomes $w + (1 - w)b$).

B2: ΔCoVaR of European Banks and the VaR Ranking (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Adjusted closes of five European banks in data/market: DBK.XETRA, BNP.PA, SAN.MC, INGA.AS, HSBA.LSE; same sample and returns as B1
- ▶ System of each bank: the equal-weighted portfolio of the other four; levels $\alpha = 1\%$ (about 40 tail days) and $\alpha = 5\%$ (about 198)
- ▶ For (c): the eleven international banks (the six US banks of B1 and these five), each with its own regional system

□ Tasks

- ▶ (a) Compute VaR 1%, $\hat{b}_{1\%}$, ΔCoVaR 1% with a block-bootstrap interval (blocks of 20 days, 999 resamples) and ΔCoVaR 5%; rank the banks at each level and compute the Spearman correlation of the two rankings.
- ▶ (b) At each level, test equal ΔCoVaR for the 10 pairs on joint block-bootstrap draws with the Holm correction, and for all five banks jointly with a Wald test (χ_4^2), at 5%.
- ▶ (c) Compute the Spearman correlation of VaR 1% and ΔCoVaR 1% across the eleven banks, with a 95% block-bootstrap interval (blocks of 20 days, 999 resamples).

B2: ΔCoVaR of European Banks and the VaR Ranking (2/2) [Proposed]

Solution discussed in class

Task

□ Report

- ▶ A table of (a); both rankings and their Spearman correlation; the pair counts (raw, Holm), the Wald statistics, p -values and decisions; the Spearman correlation of (c) with its interval
- ▶ Model: B1

□ Interpretation questions

- ▶ How many tail days does each level use, and what does this do to the power of the tests?
- ▶ Would you name one European bank “most systemic” from these data?
- ▶ Is the $\text{VaR}-\Delta\text{CoVaR}$ correlation different from zero? From one?

B3: SRISK as Published: LRMES by Simulation (1/2) [Solved]

Task

□ Data and assumptions

- ▶ JPMorgan (JPM.US, adjusted close) and the S&P 500 (GSPC.INDX, close), daily log returns in %, 5 January 2010 – 18 September 2026 [Brownlees & Engle \(2017\)](#)
- ▶ Crisis as in [Brownlees & Engle \(2017\)](#): arithmetic market return below $C = -10\%$ over $h = 22$ days; LRMES is the expected bank loss over the same h days in a crisis
- ▶ For (c): the six-month scenario ($h = 126$ days, $C = -40\%$), the shortcut $1 - \exp(-18 \text{MES}^2)$ of Lecture 18 and the prudential ratio $k = 8\%$

□ Tasks

- ▶ (a) Estimate a zero-mean GJR-GARCH(1,1) for each return series and a DCC(1,1) for their correlation, by two-step QML.
- ▶ (b) Simulate 200 000 paths of 22 days from resampled pairs of standardised innovations, keep the crisis paths and compute LRMES with its 90% range (5% and 95% quantiles of the loss); repeat with data up to 31 March 2020.
- ▶ (c) Repeat (b) with $h = 126$, $C = -40\%$ (400 000 paths), compare with the shortcut, and compute $L^* = 1 + (1 - k)(1 - \text{LRMES})/k$ for both.

B3: SRISK as Published: LRMES by Simulation (2/2) [Solved]

Task

□ Report

- ▶ The GJR and DCC estimates; the share of crisis paths; LRMES with its range, today and on 31 March 2020; the six-month LRMES against the shortcut; both L^*

□ Interpretation questions

- ▶ Which assumption of the shortcut does the simulation drop?
- ▶ Why does LRMES change between March 2020 and today?
- ▶ What is missing to turn LRMES into SRISK in money?

B3: SRISK as Published: LRMES by Simulation (Solution 1/2) [Solved]

- (a) Variance: $\sigma_t^2 = \omega + (\alpha + \gamma 1\{r_{t-1} < 0\})r_{t-1}^2 + \beta\sigma_{t-1}^2$; JPMorgan: $\hat{\alpha} = 0.017$, $\hat{\gamma} = 0.120$, $\hat{\beta} = 0.891$, persistence $\hat{\alpha} + \hat{\gamma}/2 + \hat{\beta} = 0.967$.
- DCC on the standardised residuals: $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$; $\hat{a} = 0.052$, $\hat{b} = 0.928$; last correlation $\rho_T = 0.53$.
- (b) Steps: draw 22 days of pairs $(\xi_{i\tau}, \epsilon_{m\tau})$, with $\xi_{i\tau} = (\epsilon_{i\tau} - \rho_\tau \epsilon_{m\tau}) / \sqrt{1 - \rho_\tau^2}$; rebuild $\epsilon_i = \rho_\tau \epsilon_{m\tau} + \sqrt{1 - \rho_\tau^2} \xi_{i\tau}$ with the simulated ρ ; run the GJR and DCC recursions from today's state; keep the paths with market return below -10% ; average the bank's arithmetic loss over them.
- Result: 1.3% of the paths are crises (about 2 687 of 200 000); LRMES = 10.6%, 90% range $[-2, 25]\%$.
- With data up to 31 March 2020 the same procedure gives LRMES = 23.7%.

B3: SRISK as Published: LRMES by Simulation (Solution 2/2) [Solved]

- (c) Shortcut: JPMorgan's $MES^{2\%} = 3.61\%$, $18 \cdot 0.0361 = 0.650$, so $LRMES = 1 - e^{-0.650} = 48\%$.
- Simulation for $h = 126$, $C = -40\%$: $LRMES = 33\%$.
- $L^* = 1 + 0.92(1 - 0.335)/0.08 = 8.7$ from the simulation against $1 + 0.92(1 - 0.478)/0.08 = 7.0$ from the shortcut.
- Interpretation: the simulation drops the constant β and the lognormal scaling; it starts from today's volatility and correlation and keeps the joint tail of the innovations.
- On 31 March 2020 volatility and correlation were at crisis levels, so a further -10% month took the bank down with it more; today the starting state is calm.
- SRISK in money also needs the book value of debt D and the market value of equity W .

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B4: SRISK of a European and a Romanian Bank [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Deutsche Bank (DBK.XETRA) with the Euro Stoxx 50 (STOXX50E.INDX) as its market; Banca Transilvania (TLV.R0) with BET (BET); sample of B3
- ▶ Models and crisis as in B3: GJR-GARCH(1,1), DCC(1,1), 200 000 paths of 22 days, $C = -10\%$
- ▶ Prudential ratio $k = 5.5\%$ [Engle, Jondeau & Rockinger \(2015\)](#), so $L^* = 1 + 0.945(1 - \text{LRMES})/0.055$

□ Tasks

- ▶ (a) Compute LRMES with its 90% range for both banks with data up to 18 September 2026 and up to 31 March 2020, and the share of crisis paths of each market.
- ▶ (b) Compute L^* from the LRMES and from both ends of its 90% range.

□ Report

- ▶ LRMES with range (both dates), crisis shares, L^* with its range, for both banks
- ▶ Model: B3

□ Interpretation questions

- ▶ Which bank is more exposed to a crisis of its own market?
- ▶ How wide is the range of L^* , and what does it mean for a supervisor?

B5: How Robust Is the Connectedness Index? [Solved]

Task

□ Data and assumptions

- ▶ Daily high and low prices of the 13 banks in data/market (six US, five European, TLV.R0, BRD.R0), 2010–2026, common days
- ▶ Parkinson volatility $\hat{\sigma}_t = \sqrt{252 (\ln H_t - \ln L_t)^2 / (4 \ln 2)}$, used in logs; order of the variables: US banks first, Romanian banks last
- ▶ Generalized connectedness Diebold & Yilmaz (2014) with the decomposition of Pesaran & Shin (1998) (A5)

□ Tasks

- ▶ (a) Estimate VAR(p) by OLS for $p = 1, \dots, 4$ and compute total connectedness C at $H = 5, 10, 20$ (12 values).
- ▶ (b) With $p = 1$, $H = 10$, compute C and the summed net position of the six US banks under the generalized decomposition and under Cholesky with US banks first and with Romanian banks first.

□ Report

- ▶ The 4×3 table and chart of (a); the three totals and three net positions of (b)

□ Interpretation questions

- ▶ Which choice moves the index most?
- ▶ Why does the Cholesky order change the net position of US banks?

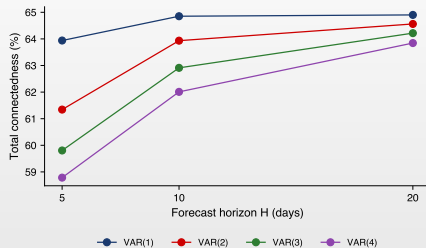
B5: How Robust Is the Connectedness Index? (Solution 1/2) [Solved]

- Steps: (1) log Parkinson volatilities; (2) $\text{VAR}(p)$ with a constant by OLS and the residual covariance $\hat{\Sigma}$; (3) Φ_h for $h < H$; (4) generalized shares, rows normalised; (5) $C = \frac{100}{13} \sum_{i \neq j} \tilde{\theta}_{ij}$.

C (%)	$H = 5$	$H = 10$	$H = 20$
VAR(1)	63.9	64.9	64.9
VAR(2)	61.3	63.9	64.6
VAR(3)	59.8	62.9	64.2
VAR(4)	58.8	62.0	63.8

- (a) The index moves between 58.8% ($p = 4, H = 5$) and 64.9% ($p = 1, H = 20$): a range of 6.1 points over all 12 choices.

B5: How Robust Is the Connectedness Index? (Solution 2/2) [Solved]



- (b) Totals: Cholesky 44.1% (US first) and 43.5% (Romanian first) against generalized 64.9%.
- Net position of the US banks: +65.9 (US first), -96.0 (Romanian first), +53.2 (generalized).
- Interpretation: the decomposition moves the index most; Cholesky gives the common shock to the variables ordered first, so the order, not the data, decides who transmits: use the generalized decomposition.

B6: Did Connectedness Change after 2020? (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Rolling total connectedness of Lecture 18: log Parkinson volatilities of the 13 banks (B5), windows of 250 days moved by 5 days, VAR(2), $H = 10$; 744 windows
- ▶ Break date: 1 January 2020; for (b) the daily series of B5 in 2010–2019 (2 373 days) and in 2020–2026 (1 593 days)

▣ Tasks

- ▶ (a) Regress the rolling index on a constant and $D_t = 1\{t \geq 2020\}$ by OLS; test H_0 : slope = 0 at 5% with ordinary and with Newey–West (50 lags) standard errors [Newey & West \(1987\)](#).
- ▶ (b) Estimate a VAR(1) separately on 2010–2019 and on 2020–2026 and compute C ($H = 10$) of each, with 95% intervals from a residual block bootstrap (blocks of 20 days, 999 draws) in each period.
- ▶ (c) Build a 95% interval for the difference $C_{2020-26} - C_{2010-19}$ from the differences of independent draws in the two periods, and decide at 5%.

B6: Did Connectedness Change after 2020? (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ The two means, the slope, both t statistics and decisions; C for both periods with intervals; the difference with its interval
- ▶ Model: B5, B9

▣ Interpretation questions

- ▶ Why are consecutive values of the rolling index nearly copies of each other?
- ▶ Why is (b)–(c) a cleaner test than (a), and what does it still assume about the break date?

B7: The FRM Penalty in One Window (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Daily returns (fractions) of the 13 banks and of the S&P 500, Euro Stoxx 50 and VIX lagged by one day (GSPC.INDX, STOXX50E.INDX, VIX.INDX)
- ▶ FRM regression [Mihoci et al. \(2020\)](#): JPMorgan on the other 12 banks and the three lagged indices (15 regressors), quantile level $\tau = 5\%$, L_1 penalty λ , windows of 63 days
- ▶ Grid of 20 values of λ log-spaced from 10^{-6} to $10^{-1.5}$; $GACV(\lambda) = \sum_t \rho_\tau(\hat{u}_t)/(n - df)$, df = number of non-zero coefficients

▣ Tasks

- ▶ (a) In the calm window ending 28 June 2019, fit the penalised quantile regression for each λ on the grid, choose the λ that minimises GACV and count the active coefficients.
- ▶ (b) Repeat (a) in the crisis window ending 31 March 2020, and list the four largest coefficients in absolute value in each window.

B7: The FRM Penalty in One Window (2/2) [Solved]

Task

□ Report

- ▶ $\hat{\lambda}$, the number of active coefficients and the four largest coefficients in each window

□ Interpretation questions

- ▶ Which banks carry JPMorgan's tail in each window?
- ▶ Why is the selected λ so small?

B7: The FRM Penalty in One Window (Solution 1/2) [Solved]

- Steps: (1) take the 63 days of the window; (2) for each λ solve the penalised quantile regression; (3) compute the residuals, df and GACV; (4) keep the λ with the smallest GACV.
- (a) Calm window (2019-03-21 to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active.
- Largest: Bank of America (0.52), Santander (0.25), Morgan Stanley (0.19), ING (-0.12).
- (b) Crisis window (2019-12-23 to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 of 15 active.
- Largest: Bank of America (0.39), Euro Stoxx 50 (t-1) (0.38), BNP Paribas (0.34), HSBC (0.32).

B7: The FRM Penalty in One Window (Solution 2/2) [Solved]

- Interpretation: Bank of America leads in both windows; in the crisis the lagged Euro Stoxx 50 and two European banks join the largest coefficients: the tail of JPMorgan becomes international.
- $\hat{\lambda}$ rises from calm to crisis, the direction the FRM reads as higher tail risk.
- λ is small because 15 regressors for 63 days make the unpenalised fit feasible: GACV prefers little shrinkage.
- The FRM is designed for many more firms than days, where the penalty is indispensable [Toma et al. \(2026\)](#).

B8: Stress Test of a Romanian Bank Portfolio (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Portfolio of 1 000 000 RON, half in Banca Transilvania (TLV.R0) and half in BRD (BRD.R0), rebalanced daily; daily log returns, days with no trades removed
- ▶ Historical windows: the bank-tax shock (10 December 2018 – 31 January 2019) and COVID-19 (15 February – 30 April 2020)
- ▶ Hypothetical scenario: weekly log returns of -25% for the Euro Stoxx 50 and -20% for BET; weekly (Friday) log returns 2010–2026 of both banks, the Euro Stoxx 50 and BET

□ Tasks

- ▶ (a) The daily log return of the rebalanced portfolio is $r_p = 100 \ln\left(\frac{1}{2}e^{r_{TLV}/100} + \frac{1}{2}e^{r_{BRD}/100}\right)$, not the average of the two log returns; in each window find the worst sum S of 10 consecutive r_p and convert it into a loss $1 - e^{S/100}$, in % and in RON.
- ▶ (b) Regress each bank's weekly return on the Euro Stoxx 50 (β_1) and BET (β_2) weekly returns by OLS; predict each bank's log return $\hat{r}_k = \beta_{1k}(-25) + \beta_{2k}(-20)$, convert it to a simple return $e^{\hat{r}_k/100} - 1$ and apply the 50/50 weights; also report the average betas and R^2 .

B8: Stress Test of a Romanian Bank Portfolio (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ Both historical losses with their dates and RON amounts; the betas, R^2 and the hypothetical loss in % and RON
- ▶ Model: A7

▣ Interpretation questions

- ▶ Which scenario hurts more: a domestic political shock or a global pandemic?
- ▶ Why is the Euro Stoxx 50 beta close to zero once BET is in the model?

B9: Which Net Spillovers Are Identified? (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Log Parkinson volatilities of the 13 banks (B5), 2010–2026; VAR order $p = 2$ chosen by BIC (up to 5 lags), $H = 10$, generalized decomposition [Diebold & Yilmaz \(2014\)](#)
- ▶ Window of the 2020 peak of the rolling index: the 250 days ending 2020-08-27, VAR(2) as in the rolling index

▣ Tasks

- ▶ (a) Build 999 bootstrap samples: resample the VAR residuals in blocks of 20 days, rebuild the series with the estimated VAR, re-estimate the VAR and recompute the connectedness table.
- ▶ (b) Report 95% percentile intervals for C and for the 13 net positions; call a bank a transmitter (receiver) only if its interval lies above (below) zero.
- ▶ (c) Repeat (a)–(b) for C in the peak window and compare the bootstrap mean with the estimate.

B9: Which Net Spillovers Are Identified? (2/2) [Solved]

Task

□ Report

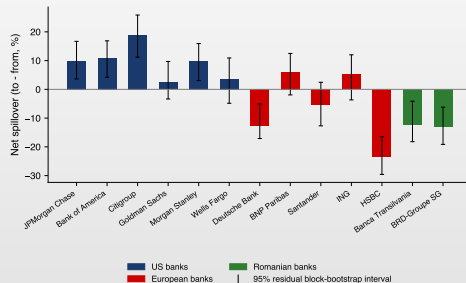
- ▶ C with its interval; a chart of the 13 net positions with intervals; the identified banks; the peak-window estimate, interval and bootstrap mean

□ Interpretation questions

- ▶ Which “transmitter” and “receiver” labels survive?
- ▶ Why is the bootstrap mean of C in the short window different from the estimate?

B9: Which Net Spillovers Are Identified? (Solution 1/2) [Solved]

- Steps: (1) VAR(2) by OLS, residuals $\hat{\varepsilon}_t$; (2) resample 20-day blocks of $\hat{\varepsilon}_t$; (3) rebuild $y_t^* = \hat{c} + \hat{A}y_{t-1}^* + \varepsilon_t^*$; (4) re-estimate, recompute the table; (5) percentiles over 999 draws.



- Net positions with 95% intervals; a label is identified only if its interval excludes zero.

B9: Which Net Spillovers Are Identified? (Solution 2/2) [Solved]

- (a)–(b) Full sample: $C = 63.9\%$, interval $[59.9, 65.8]$.
- 8 of 13 net positions exclude zero: JPMorgan, Bank of America, Citigroup, Morgan Stanley, Deutsche Bank, HSBC, Banca Transilvania, BRD-Groupe SG; the other five labels are not identified.
- (c) 2020 peak (250 days to 2020-08-27): $C = 79.3\%$, interval $[65.9, 83.0]$, bootstrap mean 75.8% , i.e. -3.5 points from the estimate.
- Interpretation: in short windows OLS noise creates spurious cross-links, so re-estimated tables drift away from the estimate: the index is biased as well as noisy [Demirer et al. \(2018\)](#).
- Report the interval next to every net number, and draw only identified labels on a network chart.

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B10: Backtesting the Girardi–Ergün CoVaR (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ The eleven international banks of B1–B2 with their regional systems; daily returns and sample as in B1
- ▶ Each day $t \geq 1\,000$, using all earlier days: $\hat{q}_t$ is the empirical 5% quantile of X_i ; $\text{CoVaR}_t^{\leq}$ is minus the empirical 5% quantile of X^{sys} on the past days with $X_i \leq \hat{q}_t$ (historical version of A2) [Girardi & Ergün \(2013\)](#)
- ▶ Joint hit $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq \hat{q}_t\}$; under H_0 , $P(I_t = 1) = \alpha^2 = 0.25\%$

□ Tasks

- ▶ (a) Count the joint hits of each bank and test $H_0 : P(I_t = 1) = 0.25\%$ at 5% with the Kupiec test [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#): $LR = -2[\ln L(0.25\%) - \ln L(\hat{\pi})] \sim \chi_1^2$.
- ▶ (b) Find the rejection region of the test in terms of the number of hits, and compute its power, assuming independent Bernoulli hits: the probability of rejecting when the true joint rate is 0.5%.

B10: Backtesting the Girardi–Ergün CoVaR (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ The number of forecast days; expected and observed hits; LR and p for JPMorgan; the range of hits and the smallest p over the eleven banks; the rejection region and the power
- ▶ Model: B1 (tests), A2 ($\text{CoVaR}^{\leq}$)

▣ Interpretation questions

- ▶ Why is α^2 the right null rate?
- ▶ What does a non-rejection tell you here?

C1: How Exposed Are Romanian Banks to Euro-Area Banking Stress? (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Banca Transilvania (TLV.R0) and BRD (BRD.R0); European system: equal-weighted portfolio of Deutsche Bank, BNP Paribas, Santander, ING and HSBC; BET (BET)
- ▶ Sample: 5 January 2010 – 18 September 2026, 3 960 trading days common to all 13 banks and the four indices (prices joined first, then daily log returns in %)
- ▶ Exposure CoVaR: quantile regression of the Romanian bank on the European system, so ΔCoVaR is the extra loss of the Romanian bank when the European system is in distress (the reverse direction of B1)

□ Tasks

- ▶ (a) Estimate the exposure ΔCoVaR of each Romanian bank at 5% and at 1% with block-bootstrap intervals (blocks of 20 days, 999 resamples), and the same measure with BET in place of the European system.
- ▶ (b) Split the sample into 2010–2017 and 2018–2026, estimate ΔCoVaR 5% in each, and test the change at 5% with $z = (\hat{\Delta}_2 - \hat{\Delta}_1) / \sqrt{\text{se}_1^2 + \text{se}_2^2}$ from independent block-bootstrap draws (or a dominance test [Castro & Ferrari \(2014\)](#)), not by overlapping intervals.
- ▶ (c) With rolling Diebold–Yilmaz tables (log Parkinson volatilities of the 13 banks, windows of 250 days moved by 5 days, $\text{VAR}(2)$, $H = 10$), compute the share of the Romanian banks' forecast error variance coming from European and from US banks.

C1: How Exposed Are Romanian Banks to Euro-Area Banking Stress? (2/2) [Proposed]

Solution discussed in class

Task

☐ Report

- ▶ Deliverable: a Quantlet folder and three slides: the numbers of (a)–(c) with intervals and tests, the interpretation, the limitations

☐ Interpretation question

- ▶ Are Romanian banks exposed mainly to euro-area banking stress or to domestic shocks, and has the exposure grown?

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- ▣ **Student prompt:** "Bank i and the system are jointly Normal with zero means: $\sigma_i = 2.5\%$, $\sigma_{sys} = 1.2\%$, $\rho = 0.5$. Compute CoVaR 1%, ΔCoVaR 1%, MES 5% (the system is the market) and SRISK with $W = 100$, $D = 1100$, $k = 8\%$, LRMES = 40%."
 - ▶ W : market value of equity; D : book value of debt; k : prudential capital ratio
- ▣ **Answer of the AI assistant**
 - ▶ $\text{VaR}_i = 2.5 \cdot 2.326 = 5.82\%$; $X^{sys} \mid X_i = x \sim N(0.24x, 1.04^2)$
 - ▶ CoVaR conditions on the bank at its VaR, $X_i = \text{VaR}_i = 5.82$: the 1% quantile is $0.24 \cdot 5.82 - 1.04 \cdot 2.326 = -1.02$, so CoVaR = 1.02%
 - ▶ Median state: $1.04 \cdot 2.326 = 2.42\%$; $\Delta\text{CoVaR} = 1.02 - 2.42 = -1.40\%$: the bank **reduces** systemic risk
 - ▶ MES is the expected loss of the bank in its own 5% tail: $\text{MES} = 2.5 \cdot \varphi(1.645)/0.05 = 5.16\%$
 - ▶ $\text{SRISK} = kD - (1 - \text{LRMES})W = 88 - 60 = 28$

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

- The answer on the previous slide contains **three** errors typical for systemic-risk measures; the rest is correct
 - ▶ Model: A1, A3 and the SRISK formula of Lecture 18; conventions of this seminar: α is a tail probability, VaR, CoVaR and MES are positive losses
- Your tasks
 - ▶ Find each error: which step, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a sign check, a limiting case, a closed form
 - ▶ Give the corrected numbers
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

C3: Does FRM@RO Still Lead Market Volatility? (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Case study of Lecture 18: [Pele et al. \(2023\)](#), the Financial Risk Meter of the BVB (FRM@RO): quantile LASSO at $\tau = 0.05$, the $J = 20$ largest companies, 14 macro variables, windows of 63 days, λ_j chosen by GACV
- ▶ The published daily FRM@RO series (file docs/frm.csv at danpele.github.io/frm_ro); the BET index (BET); for (b) and (c), the 11 BVB stocks of data/market with BET-FI, BET-NG, VIX, S&P 500, Euro Stoxx 50 and DAX
- ▶ An FRM@RO value is dated by the last day of its 63-day window; the BET volatility is the conditional standard deviation of a GARCH(1,1) with Normal innovations

▣ Tasks

- ▶ (a) Fit the GARCH(1,1) to daily BET log returns, align its volatility with FRM@RO on common days, and run Granger F tests in both directions with lags 1 to 5 on the published sample; compare your statistics with Table 3 of the paper.
- ▶ (b) Re-estimate the index with the course data from 2014 to 2026 exactly as published (quantile LASSO at $\tau = 0.05$, windows of 63 days, λ_j by GACV, average over companies), using in each window the stocks with a full 63-day history, and report its correlation with the published FRM@RO on the common period.
- ▶ (c) With your index, repeat the Granger tests of (a) on the out-of-sample period 2023–2026, and compute the influence indicator (Eq. 8) of each macro variable, to compare its ranking with Table 2 of the paper.

C3: Does FRM@RO Still Lead Market Volatility? (2/2) [Proposed]

Solution discussed in class

Task

□ Report

- ▶ The F statistics and p -values of (a) and (c) with the lag order, the correlation of (b), a table of influence indicators, one chart of the two indices with BET volatility, and the data limitation (fewer than $J = 20$ companies)
- ▶ Deliverable: a Quantlet folder and three slides: numbers, interpretation, limitations
- ▶ Model: B7, Lecture 18 (case study)

□ Interpretation question

- ▶ Does the out-of-sample evidence support FRM@RO as a leading systemic-risk signal, or can BET volatility alone explain what the index shows?

Common Pitfalls

- ▣ Putting the bank inside its own system portfolio
- ▣ Ranking banks by ΔCoVaR without intervals, or with i.i.d. intervals that ignore crisis clusters
- ▣ Using Cholesky decompositions whose conclusions depend on the order of the banks
- ▣ Testing a rolling index with ordinary standard errors
- ▣ Reading Normal-model probabilities of reverse stress scenarios as real probabilities

Take-Home Assignment (Team Project)

- ▣ Choose a financial system not covered in class (e.g. Central and Eastern European banks, insurers, crypto exchanges)
- ▣ Estimate MES, ΔCoVaR and a connectedness table, each with block-bootstrap uncertainty
- ▣ Design one historical and one reverse stress test for an equal-weighted portfolio of the system
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ If you use an AI assistant, log every error you catch in [AI_ERRORS.md](#) (as in C2)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

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